

```
ionic serve
ionic serve -port 8080 (Use
a specific port instead of the default
8100)
```

To check if your application is working successfully, open any browser and go to localhost:8080. You will see a page as shown in Figure 1.

Because they allow developers and enterprises to use a single codebase to deliver apps across various platforms, Ionic-based hybrid apps are a viable solution. They may not always equal a native program's performance and user experience, but they nonetheless manage to balance price, speed, and cross-platform compatibility. The final decision between native and hybrid development is based on the requirements, budget, and user experience that you want for your project. **END** 🐧

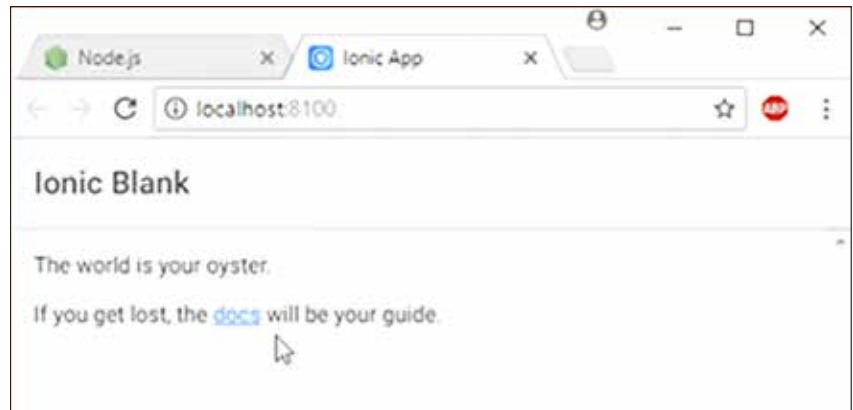


Figure 1: Ionic application

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Benefits of open source IoT middleware

Open source IoT middleware scores over proprietary middleware in many ways.

- It is easy to upgrade to new technologies with open source middleware.
- This middleware has the ability to connect with upcoming device protocols and backend applications.
- Open source middleware adoption lowers overall software cost.
- Microservices-based architecture, built using open source technologies, results in high-performance, scalability, and fault tolerance.
- It provides multi-protocol support and is hardware-agnostic.
- It offers the flexibility to work with different cloud services providers.

Middleware is necessary for managing heterogeneous devices in the IoT network. Since several IoT middleware are available in the market, selecting the right one is important. To do that, we must analyse

the business problem and arrive at the solution. We must then break the solution into services and understand the middleware needs of these. This will help us to narrow down the middleware choices. **END** 🐧

Acknowledgements

The author would like to thank Tanay Srivastava, director, Tricon Solutions LLC, for giving the required time and support in many ways while this article was being written as part of technical services efforts.

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